



Ryan Zernach Full-Stack AI Systems Engineer • Agentic Infrastructure & Product Engineering

Portfolio: [Ryan.Zernach.com](https://ryan.zernach.com) • Email: Ryan@Zernach.com • [LinkedIn](#) • [GitHub](#) • [Remote](#) • [Healthcare](#) • [FinTech](#)

HIGHLIGHTS

- Distributed systems engineer with 10+ years of engineering Payments, Agents, Mobile Apps, Websites, DevOps + MLOps Systems
- Scaled CellarChat™ tool-calling agent to 1M+ active members, complete with RLHF & CI/CD LLM evaluation scoring pipelines
- Engineered Apple Pay subscriptions to lift recurring revenue by 25%+ which lifted mobile app ratings to 4.9★ (10k+ reviews)
- Completed Gauntlet AI engineering fellowship with 2% acceptance rate and mastered the art of agentic AI orchestration
- Designed observable backend systems that support autonomous AI-powered VoIP eCommerce payment automations
- Awarded Top 10 of 10,000 international projects in 2024 NASA Space Apps hackathon with AI Voice Assistant

TECH STACK

Programming Languages

TypeScript, Python, Rust, Golang, Java, C#, Ruby, Swift, Kotlin, Bash, JavaScript, PHP, C++, Objective-C, R, Solidity, Dart, Flutter, XML

AI Systems & Machine Learning

LLMOps, vLLM, LiteLLM, RAG, Agents, MCP, Vector Embeddings, Evaluations, Semantic Kernel, LangChain, Langfuse, Computer Vision, PyTorch, Knowledge Graphs, TensorFlow, Scikit-Learn, Bedrock, Foundry

Platform & Backend Engineering

DevEx, Node.js, FastAPI, Django, Nest.js, Express, .NET Core, OpenAPI, Microservices, REST, GraphQL, WebSockets, Webhooks, gRPC, Kafka

Data & Persistence

PostgreSQL, SQL, NoSQL, Redis, Vector Databases, ETL, Snowflake, Graph Databases, Entity Relationships, Caching, BigQuery, PySpark

Cloud Infrastructure & DevOps

AWS, Azure, GCP, Cloudflare, Docker, GitHub Actions, Terraform, Kubernetes, CI/CD, CloudOps, Blue/Green, DNS, Networking

Front-End & Product Engineering

React, React Native, iOS, Android, Web, Swift, Kotlin, Flutter, Expo Router, Next.js, Redux, Streamlit, Animations, Design, User Feedback, Playwright

Distributed Systems & Operations

Queues, Redis, RabbitMQ, Analytics, Pub/Sub, Linux, Airflow, VoIP, Cron, WebRTC, Observability, OpenTelemetry, Rate Limiting, Horizontal Scaling, Machine Learning CI/CD, Event-Driven Architecture, Load Balancing

Security, Identity & Compliance

AuthN, AuthZ, OAuth, SOC 2, Identity, Role-Based Access Control, JWT, Secure Systems Design, SSO, API Security, Secrets Management

Payments, Communications & Integrations

PCI Compliance, Stripe, Apple Pay, IAP, Billing, Subscriptions, Wallets, Twilio, SMS, Push Notifications, HTTP APIs, Marketing Tech, SMTP/IMAP

Blockchain & Web3

Ethereum, Solana, dApps, WalletConnect, MetaMask, Polygon, Solidity, Smart Contracts, Ethers.js, Tokens, ERC-20, ERC-721

EXPERIENCE

Independent Full-Stack AI Systems Engineer • USA • Remote

July 2025 - Present

- Building AI-enabled products and back-end systems across fintech, semantic intelligence, and now focused on healthcare
- Productionized eCommerce platform that is powered by Twilio VoIP, realtime voice models, and DALL-E image generators
- Dockerized agentic Robotic Process Automation (RPA) via local vLLM inferences, automated web-scraping, & Kubernetes
- Published TranslateWallpaper, a cross-platform mobile app to make LLM-powered language-learning wallpapers images

Senior Full-Stack AI Systems Engineer • CellarTracker • Seattle • Remote

Aug 2023 - Jul 2025

- Scaled agentic CellarChat™ to 1M+ international members & built world's highest-rated #1 wine app
- Architected RLHF pipeline to fine-tune OpenAI's GPT-4o, OpenAPI CI/CD triggers, embeddings, & evals
- Integrated ML wine recommendations using Azure-ML + world's largest online database of wine reviews
- Maintained Python, TypeScript, React Native, C#/.NET, SQL, Redis, CI/CD, Azure AI Foundry + PromptFlow
- Helped to boost app store ratings to 4.9★ average with 10,000+ verified 5-star ratings on iOS and Android
- Embedded Apple Pay subscriptions in the React Native app to generate additional 25%+ recurring revenue
- Developed backend Python systems supporting agentic workflows, support tickets, & account management
- Improved user's experience of cellar management CRUD workflows to improve consistency and reduce taps
- Implemented PyTorch OCR image-to-text search, high-performance barcode scanning, and activity feed posts
- Streamlined the Redux store for fast, efficient, and persistent global state management between app sessions
- Rebuilt the React Native app's analytics implementations for tracking critical user interactions and in-app behavior
- Migrated 25+ years legacy support ticket system from Active Server Pages (ASP) to TypeScript Express framework
- Arranged A/B tests during paywall & onboarding experiences using phased rollout schedules for marketing analysis

Full-Stack Mobile Engineer • Archlife Industries Software • Miami • Remote	Sept 2021 – Aug 2023
• Delivered eTrading stocks/crypto exchange to compete with Robinhood: public leaderboards + copy friend's portfolio distribution • Used AWS Terraform to deploy docker container running, Java + Springboot, & NodeJS dashboards, and SQL migrations • Constructed core mobile app functionality, including: Plaid connections, trending investors, & suggested stocks • Built social network features, such as direct chat, user profiles, posts, likes, emojis, & threaded comments • Wrote the data-visualization logic for viewing and comparing the performance of different stocks/crypto • Transitioned eSports React Native codebase from JavaScript to TypeScript for long-term maintainability • Improved Ruby on Rails for push notifications, chat, & native calendar sync to improve user satisfaction • Remedied bluetooth connection & map performance for Swift iOS/iPad telecommunications app	
Data Science + Machine Learning Tutor • edX • Vanderbilt, UC Berkeley, +24 others • Remote	July 2020 – Sept 2021
• During pandemic, taught 900+ online college students Python, PyTorch, APIs, machine learning, TypeScript, SQL + databases • Managed busy schedule of remotely teaching 30+ university students per week in 1:1 setting (one hour per person) • Dozens of universities were subscribed to my employer's curriculum and online tutoring resources for students	
React Web Developer + Landscape Architect • Georgia Hybrid Landscaping • Alpharetta, Georgia	Feb 2013 – Jan 2019
• Engineered uploading videos/images, survey questions, pricing proposal, & processing payments via Java + Spring Boot • Migrated websites from PHP to NodeJS React, which successfully processed \$500k+ gross sales in three years • Managed online eCommerce payments, visitor analytics, Google AdSpend, and email marketing campaigns • Configured automated, outbound marketing email campaign schedulers & listeners using IMAP/POP • Setup observability dashboards for payments, marketing, analytics, API logs, & admin	
Payments Systems Intern (Card Processing) • Cost Plus Processing • Alpharetta, Georgia	Sept 2012 – Jan 2013
• Worked with merchant processing systems & payment gateways to understand PCI compliance and financial flows • Analyzed end-to-end card transaction lifecycles (interchange rates, risk) across critical payment networks • Built foundational knowledge of secure payment infrastructure, interchange fees, & transaction routing	

EDUCATION & TRAINING

AI Engineering Fellowship (G3) • Gauntlet AI • Remote • Austin, TX	2025
• Mastered how to command agents, automate workflows, and 100x enterprise-grade software engineering • Harnessed the power of RAG, multi-agent systems, OpenAPI, and MCP to build agentic automations	
Data Science & Machine Learning Bootcamp • Lambda School • Y Combinator • Remote	2020
• Recommendation Engines, Computer Vision (CV) Models, Time-Series Forecasting, Natural-Language Processing • Python, SciKit-Learn, PySpark, Django, Flask, FastAPI, Data Engineering, PostgreSQL Databases, Back-End APIs	
Turfgrass & Landscape Management • University of Georgia (NALP) • Athens, Georgia	2015
• Design Principles, Science of Grass, Outdoor Business Management, Landscape Architecture	
Computer Science & Mathematics Coursework • Georgia Southern University • Statesboro, Georgia	2012
• Visual Basic, Calculus, C, C++, R, Statistics, Data Structures, Programming Principles, Linear Regression	

CERTIFICATIONS

2025 • Gauntlet AI Engineering Fellowship (G3)	2020 • AutoML Pipelines (DataRobot Academy)
2024 • NASA Space Apps Hackathon Finalist	2019 • Google Cloud Platform GCP Developer
2022 • Blockchain Hackathon Participant	2019 • AWS Microservices Cloud Practitioner
2020 • TensorFlow Computer Vision	2019 • Y Combinator Startup School